Highlights

- A collaborative multi-agent semantic grasping framework overcomes the reasoning coupling of single-agent systems and fragmentation of non-collaborative multi-agent approaches.
- A collaborative semantic disambiguation strategy enables robust understanding of implicit and ambiguous natural language instructions.
- An interpretable collaborative grasp evaluation mechanism improves decision transparency and execution reliability in complex scenarios.

AgentsGrasp: A Multi-Agent Cooperative Reasoning Architecture for Semantic Scene Understanding and Grasp Planning

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Abstract

Most existing semantic grasping systems are built upon single-agent architectures, where language understanding, perception, and grasp decision-making are tightly coupled within a unified model. Although effective in constrained settings, such designs suffer from limited reasoning capacity when handling ambiguous instructions, reduced robustness in cluttered or occluded scenes, and poor interpretability of grasp decisions. These limitations fundamentally stem from the absence of explicit reasoning decomposition and information interaction.

Recent multi-agent approaches attempt to alleviate these issues by decomposing the task into multiple agents. However, in most cases, agents operate in a loosely coupled or sequential manner, lacking structured collaboration. Without effective inter-agent coordination, semantic understanding, perception reasoning, and action planning remain weakly aligned, leading to fragmented decision processes and marginal performance improvements over single-agent baselines.

To address these shortcomings, this paper proposes AgentsGrasp, a collaborative multi-agent reasoning framework for semantic grasping. Instead of emphasizing agent specialization, the framework focuses on explicit collaboration through iterative in-

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formation alignment and joint reasoning. By enabling agents to share intermediate semantic representations and continuously refine their understanding through coordination, AgentsGrasp establishes a coherent end-to-end reasoning pipeline from natural language instructions to grasp execution.

Furthermore, the framework incorporates an interpretable grasp evaluation strategy that jointly considers task constraints and execution feasibility, ensuring transparent and reliable decision-making. Extensive experiments in simulation and on a real 6-DOF robotic manipulator demonstrate that, compared with single-agent systems and non-collaborative multi-agent baselines, AgentsGrasp achieves significant improvements in grasp success rate (up to 15.3%), robustness in complex scenes, and adaptability to diverse semantic instructions. These results highlight the critical role of structured collaboration in multi-agent semantic grasping.

Keywords: Robot Grasping, Multi-Agent Systems, Implicit Instruction Parsing, Interpretable Decision

1. Introduction

Large language models (LLMs) and vision-language models (VLMs) have recently demonstrated strong semantic understanding and task-oriented reasoning capabilities in robotic applications, enabling natural language control and perception-driven decision-making in complex operational settings [? ? ? ? ?]. By leveraging commonsense reasoning, task structure modeling, and contextual understanding, these models have been successfully applied to task planning [?], navigation [?], and language-conditioned manipulation. In particular, large vision-language models (LVLMs) directly integrate visual information into linguistic reasoning pipelines, providing preliminary open-world grasping capabilities by jointly reasoning over visual observations and semantic instructions [?].

Building upon these advances, a growing body of work explores the integration of LLMs into embodied grasping and manipulation tasks. Many approaches adopt zero-shot or instruction-driven paradigms, in which language models generate high-level action sequences from natural language commands and invoke underlying execution modules to complete grasping operations [**?** **?**]. To further enhance perceptual grounding, LVLMs incorporate image inputs into the reasoning process, forming unified vision-language representations that enable object attribute perception and semantic alignment between language and action [**?** **?**]. These studies demonstrate that, under appropriate prompts and contexts, LVLMs exhibit a certain degree of visual grounding and joint reasoning ability for grasp target selection and action planning [**?** **?**].

Despite these promising results, most existing large-model-based grasping systems remain fundamentally constrained by single-agent reasoning architectures. In such settings, a single model is responsible for perception interpretation, semantic understanding, task planning, and grasp decision-making simultaneously. This tightly coupled reasoning paradigm leads to several structural limitations. First, LLMs lack native embodiment awareness and struggle to model physical properties such as 3D geometry, contact stability, and interaction constraints, even when augmented with visual inputs [**?**]. Second, LVLM-based reasoning is highly sensitive to prompt formulation and contextual priors; ambiguous or incomplete instructions often induce reasoning bias and unstable grasp strategies [**?**]. More critically, concentrating multiple reasoning objectives within a single agent results in long reasoning chains, severe functional entanglement, and limited error isolation, making it difficult to scale to multi-step grasping tasks in cluttered and dynamic environments [**?**].

To mitigate the coupling issue of single-agent systems, recent studies have introduced multi-agent formulations for robotic reasoning. However, most existing multi-agent grasping frameworks adopt loosely connected or sequential agent designs, in which agents operate independently or exchange limited information. The lack of explicit collaboration and structured information alignment often leads to fragmented reasoning outcomes, semantic inconsistencies across modules, and marginal improvements over single-agent baselines. Consequently, simply increasing the number of agents does not fundamentally address the "strong semantic understanding but weak execution" problem in language-driven grasping systems [**?**].

Motivated by these challenges, this paper proposes AgentsGrasp, a collaborative multi-agent reasoning framework for semantic grasping. Rather than emphasizing agent specialization, the framework centers on explicit collaboration mechanisms, enabling agents to iteratively align intermediate semantic representations and jointly refine grasp decisions. Through structured information sharing and coordinated reasoning, AgentsGrasp decomposes the end-to-end "language-to-action" process while preserving semantic consistency across perception, reasoning, and execution stages.

As illustrated in Fig. 1, compared with a single-agent system, AgentsGrasp effectively mitigates information interference and decision bias by distributing reasoning responsibilities and enforcing semantic alignment through collaboration. This design allows the system to maintain coherent task understanding and stable grasp execution in complex scenes, demonstrating the effectiveness of collaborative reasoning for language-driven robotic grasping.

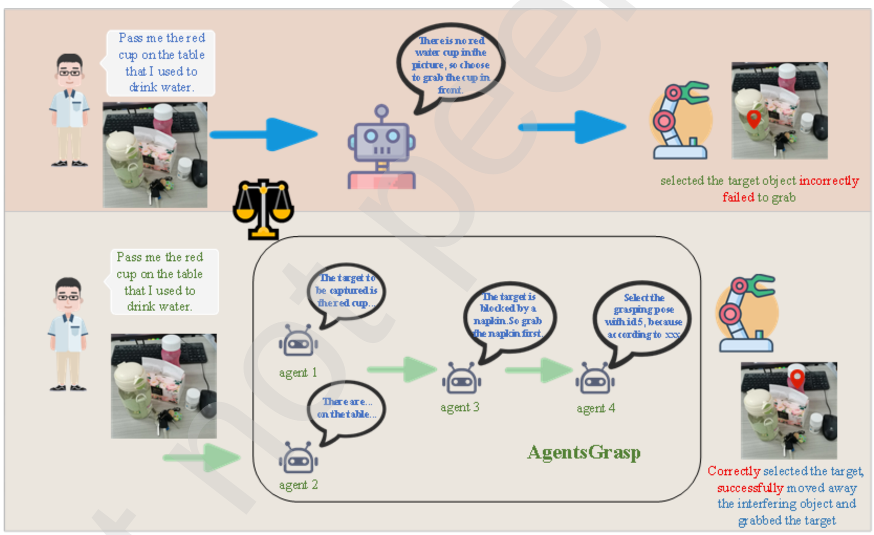

Figure 1: compares a single agent with AgentsGrasp in the same grasping task. The single agent failed due to limited information processing, while AgentsGrasp successfully completed the task through multi-agent collaboration.

The main contributions of this paper are as follows:

1.Collaborative multi-agent semantic grasping framework. We propose AgentsGrasp, a collaborative multi-agent reasoning framework that explicitly addresses the semantic interference and scalability limitations of single-agent systems and the fragmented reasoning of non-collaborative multi-agent approaches, enabling consistent end-to-end reasoning from language instructions to grasp execution.

2.Collaborative disambiguation of implicit language instructions. Unlike prior one-pass or prompt-sensitive parsing methods, the proposed framework resolves ambiguous and implicit instructions through multi-step collaborative semantic alignment, allowing robust target localization and improved generalization across diverse natural language expressions.

3.Interpretable collaborative grasp decision-making. We introduce a collaborative grasp evaluation mechanism that integrates semantic intent with execution feasibility in a structured and interpretable manner, producing transparent and controllable grasp decisions that are difficult to achieve with black-box single-agent models.

2. Related work

In recent years, with the development of large-scale pre-trained models, vision-language models (VLMs)have gained widespread attention in the field of robotic manipulation. Traditional robotic grasping and manipulation methods typically rely on explicit semantic labels, manual feature engineering, and precise scene analysis. While these methods perform well in closed tasks, they struggle with generalization under open instructions and unknown scenarios. In contrast, VLMs integrate visual and linguistic modal information, enabling robots to understand intentions from natural language, infer task objectives, and generate operational plans—thereby achieving more versatile grasping and manipulation. For instance, models like PaLM-E[**?**] and Say-Can[**?**] combine large language models (LLMs)with low-level control modules to achieve language-driven task decomposition and action planning, allowing robots to perform complex operations based on high-level semantic instructions. Recent work such as RoboFlamingo [**?**] further explores frameworks for adapting open-source multimodal models to robotic perception and control, enhancing system generalization and scalability. Meanwhile, researchers have applied VLMs to task-oriented operations[**?**] and complex scene manipulation tasks[**?**], enabling robots to understand contextual semantic relationships, infer operational objectives, and autonomously generate action sequences.

However, while VLMs demonstrate significant potential in semantic understanding and instruction reasoning, existing approaches predominantly rely on static image inputs or single-frame features, lacking the capability to model temporal dynamics, action feedback, and multi-step interactions[**?**]. These limitations restrict the models' performance in handling dynamic environments, continuous grasping tasks, and multi-stage operations.

As robot tasks grow increasingly complex, single-model systems have increasingly revealed performance bottlenecks and limited generalization capabilities in perception, reasoning, and execution. To address this, researchers have introduced multi-agent collaborative architectures[**?**], enabling complementary functions and enhanced system capabilities through coordinated cooperation among agents[**? ? ?**]. This approach has gained significant traction in recent robotics research, demonstrating remarkable potential in multimodal perception, environmental understanding, path planning, and complex operations. Particularly in manipulation and grasping tasks, multi-agent methods excel in task decomposition, semantic understanding, and efficient planning decisions. For instance, the GraspMAS system[**?**] effectively decouples semantic reasoning from grasping control through role-based division of functional agents, achieving more stable and generalizable grasping performance in open environments[**?**]. Additionally, semantic discussions and collaborative reasoning mechanisms between agents have proven effective in enhancing system robustness and task adaptability in uncertain environments. Overall, multi-agent architectures demonstrate significant advantages in robot perception, task planning, and operational execution, with modular division of labor and collaboration substantially improving system stability and intelligent decision-making capabilities.

However, how to further improve the efficiency of information sharing among agents, reduce the computational cost of collaborative reasoning, and maintain global consistency in dynamic environments remain critical issues that require in-depth research in this field.

3D perception serves as a pivotal component in implementing robust grasping strategies, particularly critical in complex, dynamic, or uncharted environments. Traditional geometric approaches typically rely on object model matching, point cloud reconstruction, and pose fitting, yet these methods remain highly sensitive to sensor noise, occlusion, and lighting variations, making it challenging to maintain stable performance in real-world scenarios[?]. To overcome the limitations of geometric methods, deep learning-driven 3D grasping research has gained momentum. Notable approaches like GPD[?], PointNetGPD[?], and Contact-GraspNet[?] have achieved significant progress in grasping accuracy and generalization by learning pose distributions through point cloud features. The Dex-Net series further introduces large-scale synthetic datasets to train physics-based simulation grasping scoring functions, substantially enhancing cross-object and cross-scenario generalization capabilities[?].

In recent years, researchers have explored more versatile strategies such as 6-DOF GraspNet[?]and UniGrasp[?], aiming to establish a unified grasping framework across different perspectives and categories. While 3D perception methods provide robots with richer geometric and spatial structural information, their performance remains heavily dependent on high-quality depth sensors and complex point cloud preprocessing workflows. In multi-objective, occlusion-prone, or dynamic environments, 3D perception still faces challenges like insufficient robustness and limited real-time performance, which require further improvements.

3. Methodology

3.1. Agent Role Configuration

In natural language-driven robot grasping tasks, the system must integrate multi-level capabilities including semantic understanding, environmental perception, target reasoning, and action planning to handle multi-object interaction and dynamic requirements in complex scenarios. Analysis of these task characteristics reveals that a single module struggles to balance semantic parsing accuracy with perceptual decision robustness. This necessitates establishing a hierarchical agent mechanism within the system to achieve collaborative coordination.

To address this requirement, this study establishes four complementary agent roles from a functional perspective, each responsible for core tasks: language comprehension, scene analysis, occlusion detection, and grasping posture selection. These agents form a collaborative closed-loop system through functional division, enabling seamless information flow between the semantic and perceptual layers. As shown in figure2, the four agents demonstrate their respective responsibilities and collaborative relationships through self-introductions, laying the foundation for the subsequent system architecture design.

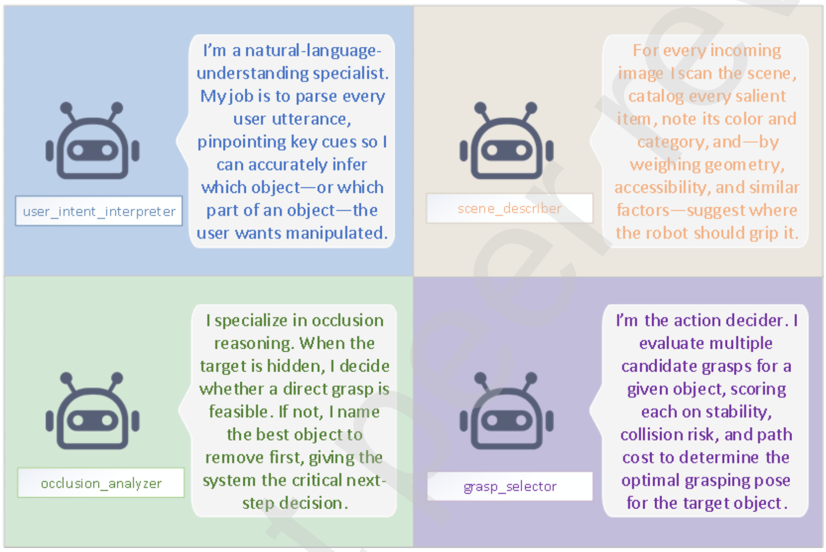

Figure 2: Role distribution of four collaborative agents in the AgentsGrasp system framework

3.1.1. user_intent_interpreter agent

Natural language instructions often contain ambiguity and context-dependent elements. For example, the command "Put my frequently used cup back in its original place" requires the robot to understand the semantic relationship between "frequently used" and "original location." The core responsibility of this agent is to extract clear task intentions and target cues from natural language inputs, generating structured semantic representations (such as target categories, attribute constraints, and action limitations). By leveraging the reasoning capabilities of large language models (LLMs), the agent can complete implicit semantics under uncertain instructions, reduce semantic-visual alignment biases, and provide high-confidence semantic priors for subsequent perception and planning modules.

The corresponding processing can be expressed as:

$$P = f_{\text{intent}}(L) \tag{1}$$

The input is the language description, the output is the language understanding module, and the output is the task intention representation.

3.1.2. scene_describer agent

The robot operating environment is often highly complex and dynamic, including multiple objects, occlusion and light change. Based on this challenge, an agent for scene understanding is designed, whose task is to understand and analyze the input image, output the objects to be grabbed in the scene, and give suggestions for grabbing these objects respectively. Specifically, the system first performs high-level semantic perception on RGB-D images to identify object categories, spatial relationships, and grasping potential within the scene.

This process can be represented as:

$$O = f_{\text{scene}}(I, D) \tag{2}$$

It represents the set of candidate objects, each containing a category, center position, and suggested grasping position.

3.1.3. occlusion_analyzer agent

In cluttered environments, over 50% of failed grasping attempts result from occlusion or object overlap. The occlusion analysis agent performs comprehensive evaluation by comparing user intent analysis from the user intent understanding agent with the object set requiring grasping from the scene understanding agent. This process involves determining target accessibility, assessing occlusion levels, evaluating grasping feasibility, and generating priority processing sequences for occluded objects. For an object with depth z in the image and other objects with corresponding depths z1, z2, etc., the occlusion judgment criteria are:

$$\text{Occluded}(o_t) = f_{\text{overlap}}(O \setminus \{o_t\}, \delta_d, dt') \tag{3}$$

The overlap indicates of occluded areas on the image plane, which serves as a visual threshold. If the target is occluded, the agent also outputs whether the target can be reached by moving the occluding object and generates candidate strategies.

$$A = (a_1, a_2, \ldots, a_n) \tag{4}$$

3.1.4. occlusion_analyzer agent

After acquiring target semantics and accessibility information, the grasping pose selection agent integrates semantic constraints with geometric features, focusing on evaluating the physical feasibility of candidate grasping poses and selecting optimal strategies. The input consists of the target object and its candidate grasping pose set, with each pose generated by GraspNet and accompanied by the following evaluation metrics:

$$S \in [0, 10] \tag{5}$$

$$C \in \{\text{Low}, \text{Medium}, \text{High}\} \tag{6}$$

The final grasping posture is selected by the optimization objective function based on the motion path length.

$$g^* = \arg\max_{g_i \in G} (S_i - \lambda_1 \cdot \mathbb{I}[C_i \neq \text{Low}] - \lambda_2 \cdot L_i) \tag{7}$$

where the weight coefficient balances the success rate, security, and execution cost of the grab.

3.2. Agent Framework of the Grasp System

This paper addresses the limitations of single-model representation and control in complex semantic grasping tasks through a structured multi-agent framework (see figure3). Given an initial RGB-D observation image I (simulated as 224×224, real-world as 640×480) and a natural language command, AgentsGrasp activates four specialized agents to perform a two-phase collaborative reasoning process for target object recognition and action planning.

In the initial phase, the intent analysis agent utilizes large language models to process user natural language inputs, extracting target categories, attribute keywords, and contextual constraints to form structured target expressions. Simultaneously, the scene understanding agent reads the task intent and images output by the intent analysis agent, employing multimodal semantic alignment technology to identify all potential object candidates in the current scene while evaluating their semantic relevance to verbal instructions. Subsequently, the occlusion analysis agent assesses whether these candidate objects are obstructed and whether occlusion levels affect grasping feasibility, generating accessibility analysis. These three agents collaboratively complete the first round of discussions to achieve global consensus on candidate target objects.

Based on this consensus, the system invokes the LangSAM model to generate visual masks for target objects through language guidance, then crops the image region focused on the target. The cropped area is subsequently fed into the GraspNet model to produce a set of grasping candidate poses, each accompanied by stability scores, collision risk assessments, and path cost estimates. In the second phase, the grasping strategy agent combines the physical grasping scores from GraspNet with the task semantic context from the previous stage, conducting pose evaluation and filtering using the following heuristic scoring function:

$$\text{Score}(g_i) = \alpha \cdot S(g_i) - \beta \cdot C(g_i) - \gamma \cdot L(g_i) \tag{8}$$

where represents the stability score, the collision risk level mapping value, the end-effector motion path length, and the empirical parameter tuning weights. The posture with the highest score is ultimately selected:

$$g^* = \arg\max_{g_i \in \mathcal{G}} \text{Score}(g_i) \tag{9}$$

This posture is used to perform the grasping action. If the grasping is successful, the system enters the termination state; otherwise, AgentsGrasp will update the image and re-activate the multi-agent reasoning process to form a closed-loop grasping control strategy until the grasping is successful or the maximum number of attempts is reached.

Unlike traditional single-model predictions for grasping tasks, AgentsGrasp simulates cognitive negotiation and strategic dialogue among multiple agents in each round, resulting in a more robust, interpretable, and highly adaptable reasoning process. This architecture demonstrates superior adaptability and success rates when handling occlusions, object ambiguities, or complex instructions.

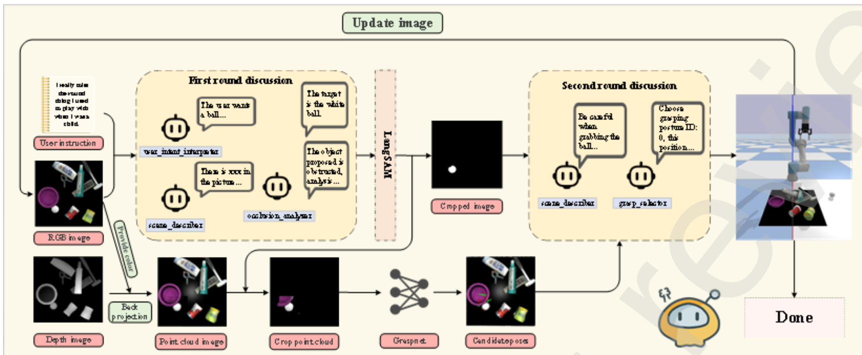

Figure 3: The overall framework of AgentsGrasp (depicting the complete process of the framework executing the grasping task

3.3. Explainable Mechanism for Agent-Grasp Pose Selection

In robotic grasping tasks, the interpretability of pose selection is critical for ensuring physical safety and strategy reliability. While the traditional deep learning-based grasping network GraspNet can directly output grasping poses, its decision-making process often remains a black box, lacking clear explanations for "why this pose was chosen."This makes it difficult to provide credible support in high-risk or complex environments.

To this end, the AgentsGrasp framework introduces an intelligent agent for grasping posture selection, aiming to achieve an interpretable and verifiable grasping decision-making process.

As shown in figure4, this visual comparison highlights the differences between black-box pose selection directly derived from GraspNet and the interpretable pose selection mechanism incorporating a pose selection agent. The upper right panel illustrates the traditional workflow: inputting a scene point cloud and directly selecting a single grasping pose through network inference, which lacks interpretability. The lower right panel demonstrates our proposed method: GraspNet first generates multiple candidate poses, followed by the pose selection agent conducting analysis and weighted screening based on stability scores, collision risks, and motion path lengths to ultimately output the optimal grasping pose. This mechanism not only preserves the semantic understanding capabilities of large models but also assigns clear physical significance to grasping decisions through explicit metrics, achieving interpretable and robust grasping strategies. The GraspNet-only approach, while providing results through direct network inference from scene point clouds, remains opaque to users regarding the decision-making process and fails to explain why specific poses are selected. In contrast, our method involves GraspNet generating multiple candidate poses, which are then analyzed and evaluated by the pose selection agent. Specifically, the system calculates each candidate pose's stability score (measuring grasping firmness after clamping), collision risk (assessing potential obstacle interference along the grasping path), and motion path length (reflecting action overhead for grasping execution). Building on this foundation, the Pose Selection Agent employs a weighted optimization strategy to identify the optimal grasping posture from a candidate pool. This approach not only preserves the semantic comprehension capabilities of large models but also provides explicit decision-making rationale through measurable metrics, thereby enhancing both interpretability and robustness of the results.

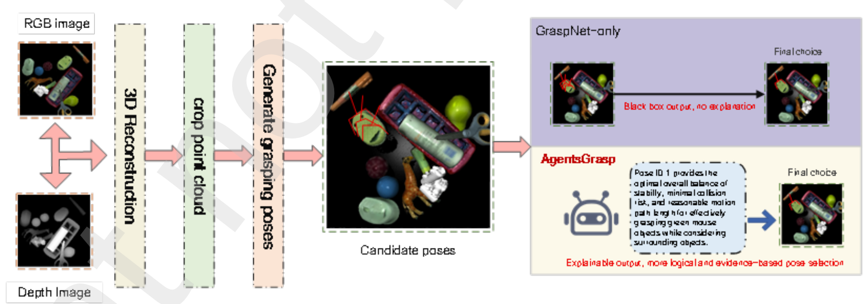

Figure 4: Comparison of GraspNet-only and the proposed method in grasping pose selection mechanism

4. Experiment

To comprehensively evaluate the effectiveness and adaptability of the proposed multi-agent semantic grasping system (AgentsGrasp), this paper designs and implements a series of experimental validations ranging from simulation to real-world environments.

4.1. Experimental Data Set and Platform Configuration

4.1.1. Simulation Data Sets and Platforms

The simulation platform experiment establishes a unified testing environment in the PyBullet simulation framework. This experimental setup comprises a UR5 robotic arm, a ROBOTIQ-85 gripper, and an Intel RealSense L515 depth camera. The system captures real-time scene images (RGB-D) and uniformly crops the original images to a 224×224 resolution for input into a multi-agent architecture to perform inference and grasping operations. All agent modules in the proposed system are implemented using Qwen2.5-VL's large-scale vision models, leveraging their powerful multimodal reasoning capabilities to accomplish tasks such as intent interpretation, scene understanding, occlusion analysis, and grasping selection.

The experiment designed task scenarios with different difficulty levels (Easy, Medium, Hard), and designed 10 representative test scenarios for each level, with a total of 30 test cases. Each scenario was repeated 15 times to ensure stability.

Table 1: Grasping-difficulty benchmark settings.

Difficulty	**Core differentiators**	**Number of test scenarios**
Easy	Few targets, minimal occlusion, and clear instructions	10
Medium	Focus on understanding implicit language instructions and handling semantic-reasoning challenges in fuzzy or omitted instructions	10
Hard	Contains numerous targets, complex occlusions, and highly similar objects, highlighting the system's adaptability to mixed environments	10

At the same time, comparative method experiments and ablation experiments are conducted in different task scenarios with different difficulty levels, aiming at the quantitative analysis of the roles of each agent module in semantic parsing, scene understanding, occlusion processing and grasping decision, and further verifying the interpretability of the grasping posture selection mechanism.

4.1.2. Real-world datasets and platforms

In real-world scenarios, an experimental platform based on a real robotic arm was developed to evaluate the language adaptability and execution performance of the AgentsGrasp framework in physical environments. The platform utilizes a six-degree-of-freedom RoARM-M1 robotic arm with its accompanying gripper, combined with an Orbbec Astra Pro Plus depth camera to capture RGB-D data of the scene. The backend system runs on a high-performance server equipped with an NVIDIA GeForce RTX 4090 (24GB VRAM), where communication and collaborative control between agent modules are achieved through ROS2. To maintain consistency with the simulation experiments, the grasping pose generation still relies on the same pre-trained GraspNet network.

In the experimental design, we implemented an Instruction Variability Adaptation test to evaluate the system's robustness and generalization capabilities across different language styles. Specifically, multiple volunteers were invited to issue grasping commands to the robot using various linguistic habits within the same task scenario, covering four typical styles as shown in Table1. The experiment designed 10 real-world scenarios, each containing 10-20 objects spanning categories like food, stationery, and daily necessities. Each language style was tested 10 times, with success rates and average action counts recorded for each category.

After receiving the command, the system is completed by the intelligent agent of language understanding, scene analysis and posture selection to complete the target recognition and grasping decision, and finally the mechanical arm performs the operation, realizing the end-to-end human-machine interaction closed loop.

Table 2: Types of natural-language instructions in the grasping benchmark.

Type of instruction	**Sample expression**	**Language feature description**
Concise	"Get bottle"	The instructions are brief and to the point, with no modifiers.
Colloquial	"Help me get that red bottle."	It has natural human-speech characteristics and contains redundant components.
Imperative	Capture the red bottle	Clear action verbs and objectives, grammatical correctness.
Implicit	I'm thirsty and need something to drink	The meaning is obscure and requires contextual reasoning.

4.1.3. Evaluation indicators

To quantify the grasping effect of different systems, the following five key indicators are used to evaluate the experiment:

(1)Average Success (average success rate): The average percentage of tasks successfully completed across all scenarios.

(2)Average Step: The average number of actions (decisions) required to complete a task. A lower value indicates higher efficiency.

(3)Average Success Step (average number of steps required to complete a task): This metric measures the compactness of high-quality paths by calculating the average number of steps needed to complete successful samples.

(4)Stability (stable grasping rate): The proportion of successfully grasped samples where the target remains stable during lifting and movement without slipping or dropping, serving as a metric for evaluating the reliability of grasping posture and force control strategies.

(5)Semantic Match (semantic match success rate): The percentage of crawled objects selected by the system that are semantically consistent with the instructions, used to evaluate the accuracy of language understanding and target selection.

4.2. Simulation Experiment

4.2.1. Verify the effectiveness of AgentsGrasp

This experimental section features three difficulty levels (Easy, Medium, Hard). The four representative scenarios (a, b, c, d) in the diagram correspond to randomly selected experimental settings. The yellow-boxed images represent Easy-level scenarios with clear instructions, while the light-blue-boxed images depict Medium-level scenarios containing ambiguous instructions. The red flag symbol in each diagram indicates the system's final grasping target. For example, in figure5's a section, the Easy-level instruction is "grasp the strawberry," whereas the Medium-level instruction states "I want something to eat." Both instruction types demonstrate the AgentsGrasp framework's ability to accurately interpret semantic intent and successfully locate targets, showcasing strong language comprehension and contextual reasoning capabilities.

As shown in figure6, for hard-scenario tasks where red markers indicate target objects, the environment contains highly similar targets and complex occlusions. Even with multiple possible interpretations of the command semantics, the system achieves robust recognition and accurate grasping through multi-agent collaboration, demonstrating its generalization and robust performance in complex tasks.

To verify the advantages of the proposed method, two types of comparison methods are set up: Single Agent System: This system directly employs the Qwen2.5-VL model for joint inference of scene input and semantic prompts, generating grasping pose recommendations without additional structural modules, equivalent to an end-to-end processing approach.

The ThinkGrasp Replication System: Built upon the visual-language grasping framework outlined in the paper "ThinkGrasp: A Vision-Language System for Strategic Part Grasping in Clutter"[**?**], this system replicates its core methodology by employing large models to identify and prioritize visual targets, then selecting grasping candidates through rule-based optimization strategies. To ensure comparability, the system

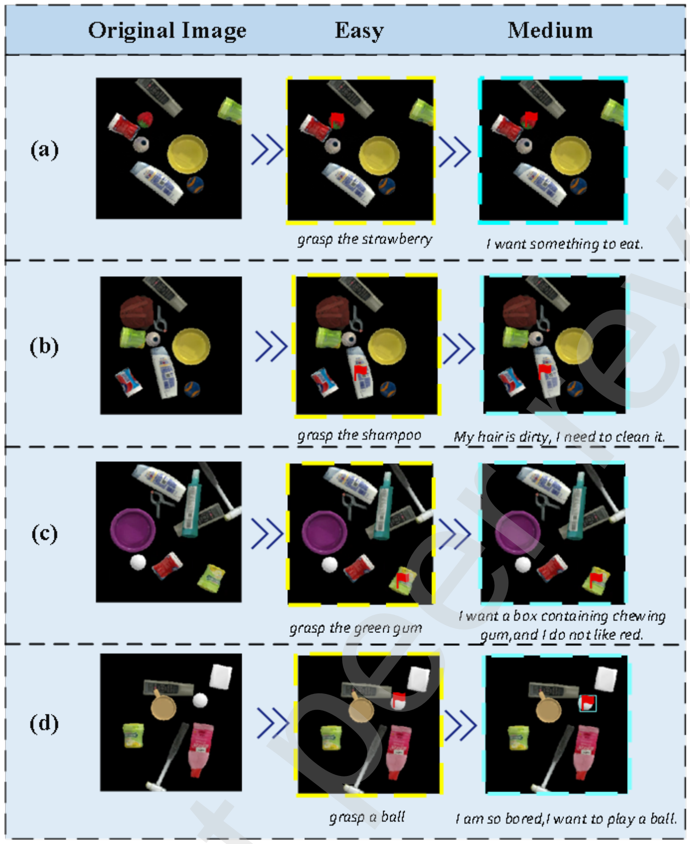

Figure 5: Experimental section display

utilizes the same visual backbone, large models, and visual-language models as the original.

GraspMAS reproduction System It is implemented based on the Multi-Agent Language-driven Grasp Detection framework proposed in the paper "GraspMAS: Zero-Shot language-driven Grasp Detection with Multi-Agent System"[**?**], and its core design idea is reproduced. This method introduces multiple agents with relatively independent functions to conduct parallel analysis and reasoning on visual perception results and language instructions, and generates candidate results for grasping under zero-shot conditions. During the reproduction process, the multi-agent collaboration paradigm of the original method is followed. Without introducing additional task decomposition or hierarchical

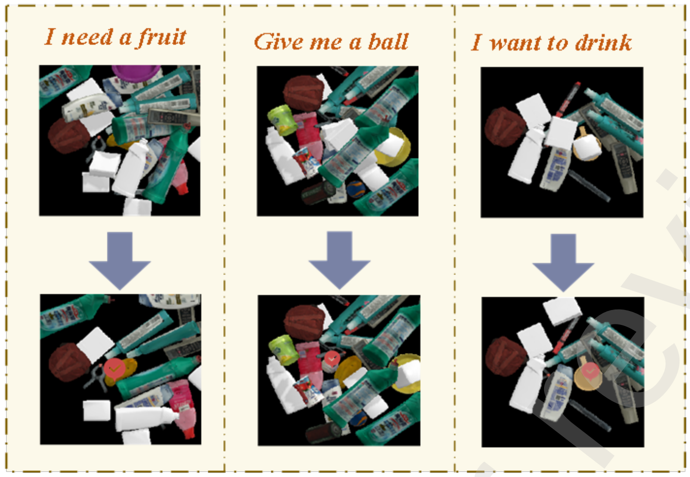

Figure 6: shows the hard scenario section

role design, the grasping decision is completed through multi-agent parallel reasoning to ensure consistency with the original method and the fairness of the comparison experiment.

The experimental results in Table 3 demonstrate that the proposed AgentsGrasp system consistently outperforms existing methods across different task difficulty levels, validating the effectiveness and adaptability of the multi-agent collaborative reasoning framework for semantic grasping tasks.

In the Easy difficulty setting, where object structures are clear and semantic instructions are explicit, all methods achieve relatively high success rates. Nevertheless, AgentsGrasp attains a 100% task success rate with an average execution step count of only 1.36, significantly outperforming the Single-Agent system (3.52 steps), ThinkGrasp (3.16 steps), and the GraspMAS baseline (2.10 steps). Although GraspMAS benefits from multi-agent parallelism and exhibits improved efficiency compared with single-agent approaches, AgentsGrasp further reduces redundant interactions through structured role assignment and collaborative reasoning. These results indicate that, under simple and well-defined scenarios, explicit inter-agent coordination enables more efficient instruction parsing and grasp execution.

In Medium difficulty tasks, the primary challenges arise from incomplete seman-

Table 3: Performance comparison across difficulty levels for four methods.

Difficulty	Metric	Single-Agent	ThinkGrasp	GraspMAS	AgentsGrasp(our)
Easy	Average Success↑	0.900	0.940	0.970	1.000
	Average Step↓	3.520	3.160	2.100	1.360
	Average Success Step↓	1.920	2.800	1.850	1.360
Medium	Average Success↑	0.850	0.850	0.920	0.980
	Average Step↓	6.200	6.320	3.480	1.640
	Average Success Step↓	3.500	3.800	2.600	1.388
Hard	Average Success↑	0.642	0.701	0.740	0.792
	Average Step↓	19.850	20.410	21.300	21.150
	Average Success Step↓	17.560	18.120	18.300	17.980

tic descriptions and ambiguous language expressions. Both the Single-Agent and ThinkGrasp systems achieve a success rate of 0.85, reflecting limited capability in handling implicit instructions and semantic ellipsis. GraspMAS improves the success rate to 0.92 by leveraging multiple agents, suggesting that multi-agent reasoning can partially alleviate semantic ambiguity. In contrast, AgentsGrasp achieves a substantially higher success rate of 0.98 while reducing the average execution steps to 1.64. This performance gain is mainly attributed to the intention interpretation agent, which provides high-level semantic guidance, as well as the collaborative information-sharing mechanism among specialized agents, effectively preventing target misjudgment and redundant grasp attempts caused by incomplete or ambiguous instructions.

In the Hard difficulty scenario, characterized by dense object distributions, severe occlusions, and similar object geometries, the overall success rates of all methods decrease compared with Easy and Medium tasks, which is consistent with expectations. AgentsGrasp still achieves the highest average success rate of 0.792, outperforming Single-Agent (0.642), ThinkGrasp (0.701), and GraspMAS (0.740). In terms of efficiency-related metrics, including average execution steps and average successful grasping steps, AgentsGrasp exhibits performance comparable to GraspMAS and

ThinkGrasp. This suggests that under extremely complex conditions, the system prioritizes robustness and decision reliability over aggressive step minimization to ensure task completion.

Overall, these results indicate that while GraspMAS demonstrates the effectiveness of multi-agent cooperation for language-driven grasping, the hierarchical organization and intention-guided collaborative reasoning adopted in AgentsGrasp enable more reliable semantic understanding and decision-making, leading to consistently higher success rates across varying task complexities.

4.2.2. Multi-agent Ablation Experiment

To thoroughly analyze the roles of each agent module (intention analysis Agent, scene understanding Agent, occlusion analysis Agent, and grasping decision Agent) in the AgentsGrasp system during grasping tasks, this study conducted systematic ablation experiments by sequentially isolating key modules to evaluate their impact on overall performance. Through comparison with the complete system, we quantified each sub-module's contribution to metrics like success rate and inference efficiency across different task difficulty levels. The research designed four ablation configurations to verify the individual contributions of each agent module to the system's grasping performance. Under the same three difficulty levels (Easy/Medium/Hard) as the main experiment, the following four ablation configurations were tested and compared with the complete AgentsGrasp system:

Configure A1 (w/o Intent Agent): In this configuration, the system removes the intent parsing agent module. The user's language commands are no longer processed through a dedicated intent understanding process but are directly passed to the subsequent scenario understanding module. This setting is used to evaluate the impact of intent understanding on semantic-guided grasping.

Configuration A2 (w/o Scene Agent): In this configuration, the system omits the scene understanding agent module, meaning it neither performs semantic interpretation on input images nor conducts semantic localization or region perception analysis of target objects. The grasping strategy directly infers actions based on user intent and initial visual input, bypassing the environmental semantic parsing process. This ablation setting aims to evaluate the contribution of scene understanding capabilities to overall grasping performance, with a focus on its impact on grasping success rate, target recognition accuracy, and strategy rationality. A3 configuration (without Occlu-

Table 4: Performance comparison across difficulty levels.

Metric	A1	A2	A3	A4	AgentsGrasp
Easy					
Average Success	0.960	0.880	0.985	0.945	1.000
Average Step	1.800	2.400	1.650	1.500	1.360
Average Success Step	1.700	2.100	1.620	1.440	1.360
Medium					
Average Success	0.900	0.800	0.930	0.920	0.980
Average Step	2.100	3.500	1.950	1.900	1.640
Average Success Step	1.890	2.800	1.800	1.750	1.388
Hard					
Average Success	0.650	0.450	0.520	0.600	0.813
Average Step	22.050	29.600	25.460	23.218	19.129
Average Success Step	18.500	24.000	22.500	19.600	16.912

sion Agent): This setup removes the occlusion analysis module, making it impossible to evaluate or infer target accessibility. The experiment aims to assess how occlusion processing affects grasping feasibility in complex scenarios.

A4 (w/o Selection Agent) configuration: This setup removes the frame-by-frame semantic reasoning and ranking mechanism for candidate grasping poses from the selection agent. Instead of relying on large models to subjectively evaluate grasping poses through language-based semantic understanding, the system directly employs candidate grasping poses generated by GraspNet for execution. This configuration is designed to investigate the impact of semantic-guided grasping selection strategies on task completion quality.

The above four sets of ablation Settings were compared under the same 30 test scenario instructions as the main experiment, and each scenario was repeated 15 times, so as to quantitatively measure the specific contribution of each agent module to the system performance. The statistical indicators are shown in Table4.

4.3. Real-world experiments

This section features real-world experiments to evaluate the AgentsGrasp framework's language adaptation capabilities in physical environments.

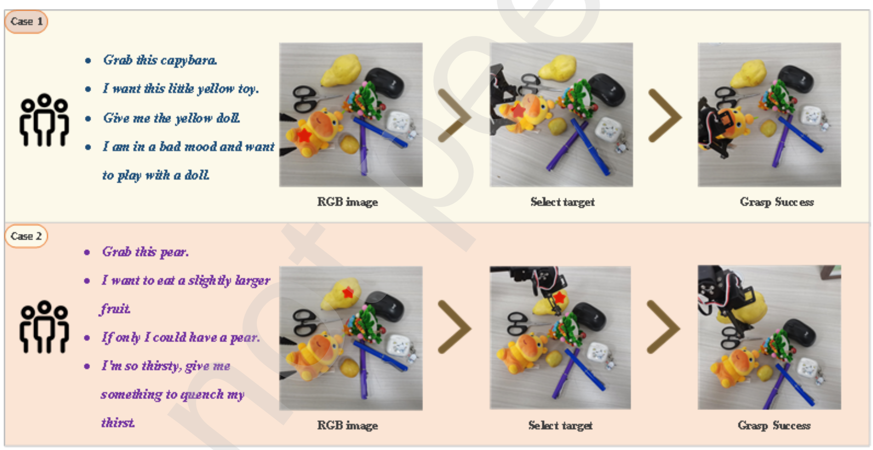

Figure 7: Simple real-world experiment

As shown in figure7, the diagram presents two real-world cases where the red five-pointed star denotes the system-identified grasping target. Under identical scenarios and targets, volunteers delivered grasping commands to the robot in four distinct linguistic styles: concise, conversational, imperative, and implicit.

figure8 further illustrates three real-world scenarios with higher complexity. Compared to figure7, these scenarios feature more objects, stronger occlusion, and background interference. Under identical target conditions, volunteers input instructions in different language styles. This experimental group systematically evaluates AgentsGrasp's comprehensive adaptability and grasping stability under semantic ambiguity, visual interference, and linguistic diversity. Experimental results demonstrate that

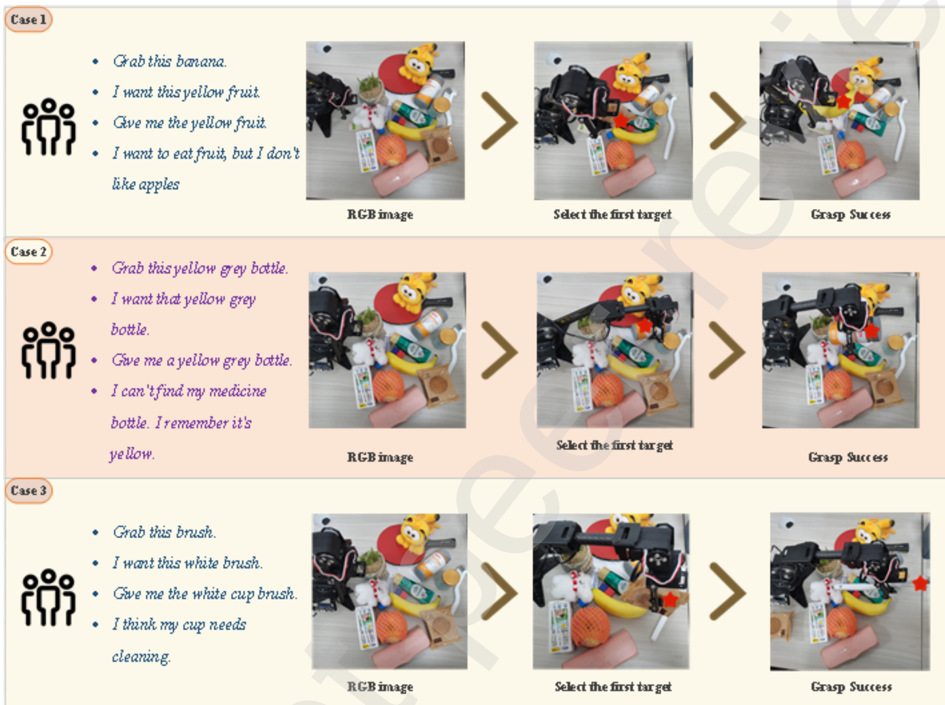

Figure 8: A more complex real-world experiment

AgentsGrasp can accurately interpret user intent and select targets even under complex visual interference and semantic ambiguity. For instance, in the scenario with the command "My task is to make my cup clear" (Case 3 in figure7), the agent automatically focuses on objects resembling cups or those used for cleaning. By leveraging spatial relationships identified through scene understanding, it determines the priority object for grasping and generates a logical grasping posture.

In contrast, the traditional single agent method often fails to analyze the situation of oral or implicit instructions comprehensively, and can only successfully capture the situation when the sentence is clear.

As shown in Table5, AgentsGrasp significantly outperforms the Single-agent method

Table 5: Performance comparison across instruction styles (Average %).

Instruction Style	Method	Average Success (%)	Stability (%)	Semantic Match (%)
Simple	Single-agent	78.5	82.0	85.3
	AgentsGrasp	88.2	90.5	92.4
Anglicized	Single-agent	74.1	79.8	80.6
	AgentsGrasp	85.6	88.9	90.3
Command	Single-agent	80.2	84.0	86.7
	AgentsGrasp	89.4	91.7	93.5
Implicit-style	Single-agent	70.8	75.5	78.1
	AgentsGrasp	82.3	86.8	88.9

across all four instruction styles. The average success rate (Average Success) improved by 8%–12%, demonstrating that the multi-agent collaboration mechanism achieves more stable grasping tasks in complex semantic and dynamic scenarios. Regarding stability (Stability), AgentsGrasp maintains better posture stability during object lifting and movement, with significantly reduced dropping and sliding phenomena. The semantic match rate (Semantic Match) also increased by 5%–10%, indicating the system's enhanced accuracy in semantic target interpretation and action mapping under multi-style language instructions.

The performance gains are most pronounced in implicit and conversational instruction scenarios, demonstrating the advantages of multi-agent semantic collaboration in understanding and execution under weakly constrained linguistic conditions. This indicates that the proposed AgentsGrasp system exhibits enhanced robustness and generalization capabilities in real-world, diverse semantic environments.

The results demonstrate that AgentsGrasp achieves cross-scenario transfer without requiring additional annotations or model fine-tuning, showcasing excellent environmental adaptability and language robustness. This indicates that the multi-agent semantic grasping framework not only proves effective in simulation environments but also enables end-to-end natural language-driven operations on physical robotic systems, providing a directly applicable technical foundation for future human-robot collaboration tasks.

5. Conclusion

This paper proposes AgentsGrasp, a multi-agent semantic collaborative reasoning framework designed for natural language-driven robotic grasping tasks. By integrating specialized agents for language comprehension, scene analysis, occlusion detection, and grasping posture selection, the system achieves end-to-end reasoning from verbal commands to physical execution. Experimental results demonstrate that this approach outperforms existing methods in grasping success rate, operational efficiency, and adaptability to implicit instructions. Furthermore, the interpretable grasping posture evaluation mechanism significantly enhances the transparency and controllability of the robotic grasping process.

However, the system still has some limitations. The current framework is mainly designed for static desktop crawling scenarios, lacking real-time response and adaptive adjustment capabilities to dynamic environmental changes.

Future research can focus on three key directions: First, integrating multi-view perception with cross-modal feature alignment technology to enhance system positioning accuracy and robustness in complex scenarios. Second, expanding the system's task capabilities from basic grasping to include more complex operations like sorting and assembly. Finally, exploring advanced semantic collaboration mechanisms among multi-agent systems, enabling flexible decision-making and precise grasping based on semantic context when handling multiple similar targets. This approach will drive the evolution of natural language-driven robot operations toward greater intelligence and versatility.